ACCESS FORM

Software Requirements Specification

Version 1.0

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Revision History

Date (dd/mm/yyyy)	Version	Description	Author
10/11/2025	1.0	Access form is a web-based system that is used to create and manage the surveys and forms. By the help of this system, users can easily fill their surveys and admin can review users' surveys and able to approve them. The system has accessible features like screen reader support, keyboard navigation and color-blind friendly themes, so that every user can easily access to the system. Users can submit their surveys via voice. Access Form is secure and reliable with encryption login. Users can only access what their role allows and automatically save surveys to prevent of data loss. The system is fast with dashboard and load surveys in just few seconds. It is also compatible for both mobile and desktop computers. The project is designed for students and the organizations to make easy and fast to create, manage and collect the surveys and forms.	BC220203921

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SRS Document

1. Scope of Project:

Access Form is a web-based survey builder application which is made by the focusing on web accessibility. Its main purpose is to easily accessible for all types of peoples especially for the persons who have any disability like visual, hearing, movement issues, understanding problem and learning difficulties.

It must follow **Web Content Accessibility Guidelines (WCAG 2.1)** and design the system in which users, especially those who have disabilities use easily.

Three main panels are used in the system:

- **Admin Panel:** For the management of users and surveys.
- **Form Creator Panel:** Use to create, edit, publish, analyze the forms.
- **Respondent Panel:** Where Users are able to complete the surveys via web, voice.

1.1. Project Overview:

Access form is a **web-based surveys builder application**, which allows every type of users to create or attempt the surveys even they have any disabilities. It is quite different from other surveys tools like Google form, because **Accessibility Features** are especially focused.

In this application the **Form Creator** easily create surveys with the help of drag-and-drop. He can add questions and apply different accessibility settings like high-contrast theme, keyboard only navigation screen reader support and alt-text for images.

System provides multiple response moods for the users who attempt to fill the surveys like web form, voice-based responses, which make more easier to fill the survey form for all types of users. The form creator gets an easy-to-use analytical dashboard where charts and tables can also be read by screen reader. The creator can also export data in different formats like CSV, PDF or Excel.

At the end conclusion is that the application must follow **WCAG 2.1,** which make it easier to use for any types of users.

1.2. Scope Boundaries:

1.2.1. <u>What the system will do:</u>

- Drag-and-drop interface provides to form creator for survey creation.
- Accessibility features will be included like
  - Keyboard-only navigation
  - Screen reader support
  - High-contrast themes
  - Alt text for images
- Users can submit surveys responses through:
  - Web form
  - Voices
- Different Question types will be supported.
- An analytical dashboard will be provided to view results.

- Data can be exported in CSV, PDF, or Excel.

1.2.2. What the System will not do:

- It is not work like a general-purpose website builder.
- It is not able to handle financial transactions.
- The system does not include any real-time chat or support system.
- The system does not support AI-based automatic form generation.

2. Functional and Non-Functional Requirements

2.1. Functional Requirements

2.1.1. User Authentication:

- The system provides secure login/logout to the user.
- The system supports the three user roles, which are admin, form creator and respondent.

2.1.2. Admin Panel:

- Admin have rights to manage the users accounts.
- Admin can review surveys and has the authority to approve them.
- Admin can manage users' complaint related to survey.

2.1.3. Survey Creation Module:

- Support drag-and-drop functionality for survey creations.
- The system provides various questions formats, including MCQs, text based, file upload and rating scale options.
- Each question supports accessibility settings such as alt text, labels and hints.
- The system support themes such as high contrast and easy-to-read modes.

2.1.4. Accessibility Features:

- Full keyboard accessibility is provided across the system.
- The system works with screen readers so the user who cannot see well can read content.
- The system checks that all images have alt text for better accessibility.
- The system allows users to resize fonts and switch between different themes.

2.1.5. Response Collection:

- The system collects web-based survey responses.
- The system authorizes the users to give their responses through their voices. (speech-to-text)

2.1.6. Analytical Report:

- The system will provide dashboard.
- The system shows graphs and tables in a format that screen reader can read easily.

2.2. Non-Functional Requirements

2.2.1. Usability:

- Interface must be simple, clean and easy to accessible for the users.
- The system should provide color palettes that are suitable for the color-blind users.
- Users should be able to navigate the entire system using keyboard.

2.2.2. Performance:

- The Dashboard should load in under 3 Second.
- Surveys responses should be submitted in under 2 second.
- The system must be able to handle more than 100 form submissions at the same time.

2.2.3. Security:

- The system uses HTTPS to keep login information encrypted and secure.
- The system controls access based on user roles, so users can only do actions allowed for their role.
- User password are stored in encrypted form for safety.
- Only authorized user can access the admin panel.
- The system prevents unauthorized users from changing important data.

2.2.4. Compatibility:

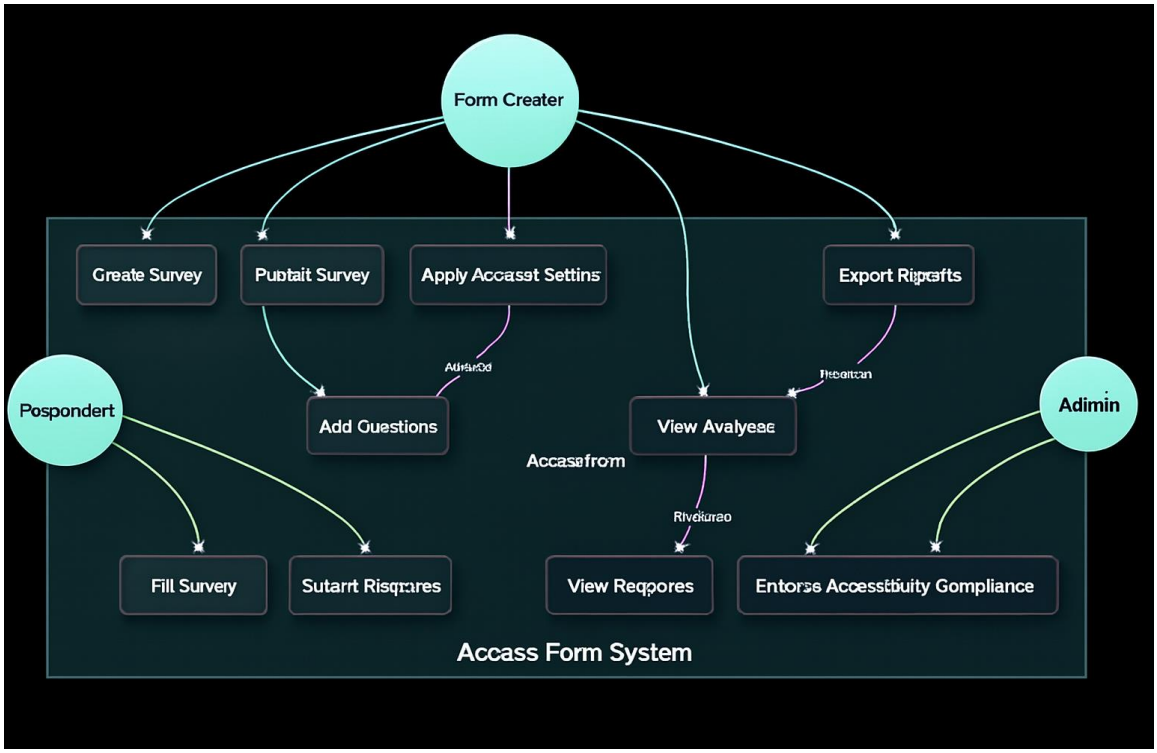
- The system must work properly on major browsers like Chrome, Firefox, Safari, and Edge.
- The system needs to run smoothly on both mobile devices and desktop computers.

2.2.5. Reliability:

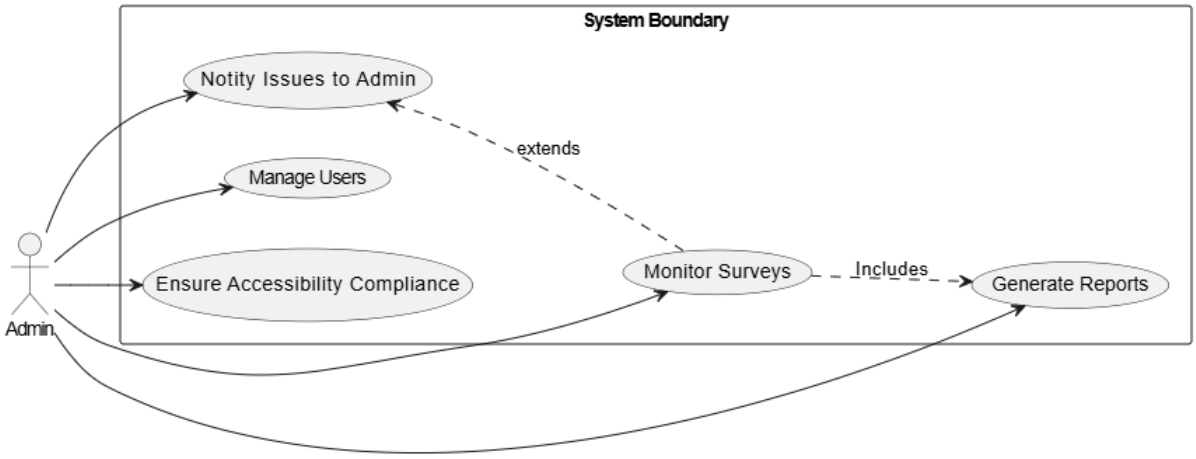
- The system must work without crashing during normal use.
- The system should stay availble so users can access it anytime.
- The system has to recover quickly if a small error happens.
- The system automatically saves surveys every 5 seconds to prevent data loss.

3. Use Case Diagram(s):

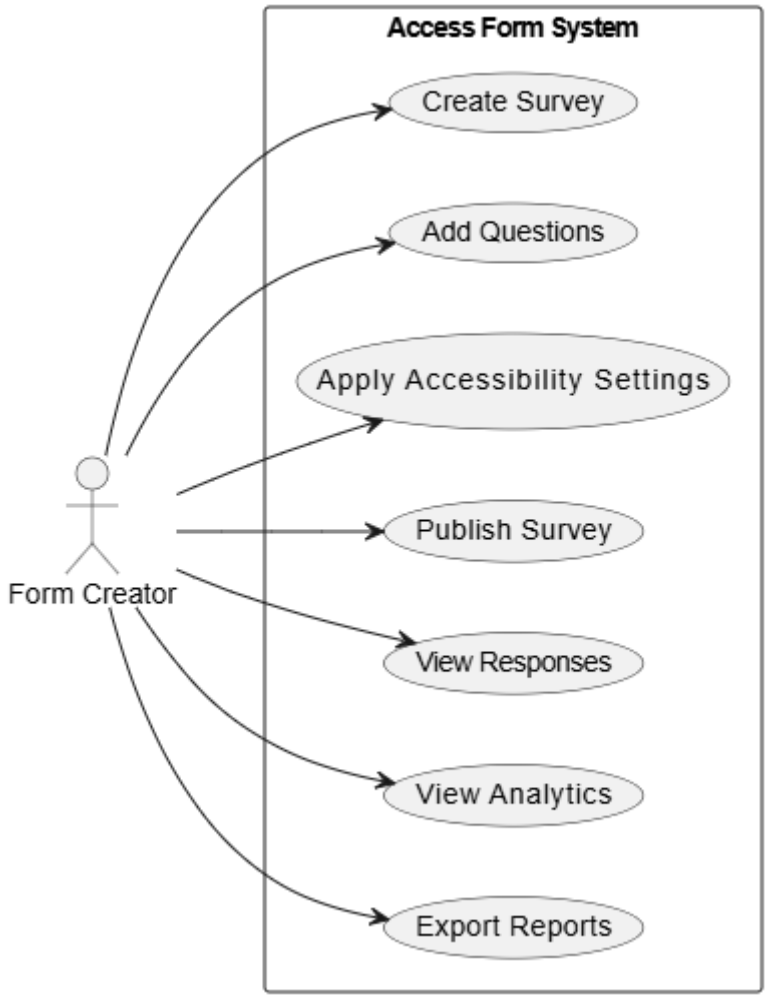
3.1. Combined Diagram: Access Form



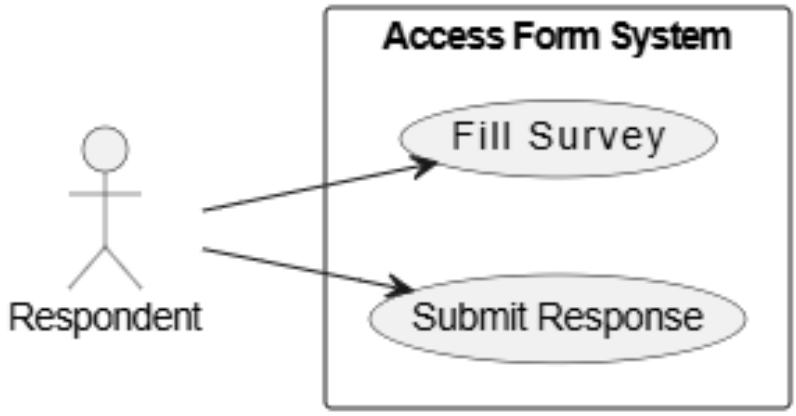
3.2. Diagram 01 – Admin Panel



3.3. Diagram 02 – Form Creator Panel



3.4. Diagram 03 – Respondent Panel



3.5. Usage Scenarios:

3.6. Admin Panel

3.6.1. Monitor Surveys

Use Case Title	Monitor Surveys
Use Case ID	UC-AP-01
Actions	• Admin Logs into the system • Open the admin panel • Select “Monitor Surveys” option • View Surveys status (Active, closed, pending) • Check response counts and performance issues.
Description	Admin checks the status of all surveys, their responses and overall progress.
Alternative Paths	• If Surveys list failed to load, system shows an error message • Admin retries or refreshes the page
Pre-Conditions	• Admin must be logged in • Surveys must exist in the system
Post-Conditions	Admin successfully views survey monitoring details.
Author	BC220203921
Exceptions	• Server error while loading surveys • Network issue prevents data loading

3.6.2. Manage Users

Use Case Title	Manage Users
Use Case ID	UC-AP-02
Actions	• Admin Select “Manage Users” • View List of system users. • Add new user or updated or delete user. • System saves changes.
Description	This use case allows the admin to manage users accounts, roles and permissions.
Alternative Paths	If user data is invalid, system shows validation message
Pre-Conditions	Admin must have authorization rights
Post-Conditions	Updated users’ information is stored in system
Author	BC220203921
Exceptions	• Duplicate user entry • System unable to save user record

3.6.3. Generate Reports

Use Case Title	Generate Reports
Use Case ID	UC-AP-03
Actions	• Admin open "Generate Reports" • Select survey or time range • Choose report type (summary, detailed) • System generates a report • Admin download or view it
Description	Admin creates summary reports for surveys, users, or system performance.
Alternative Paths	If filters are missing, system asks admin to fill required fields
Pre-Conditions	Survey data must exist
Post-Conditions	Report is generated and availble to admin
Author	BC220203921
Exceptions	• Report generation fails due to server issue • No data availble for selected filters

3.6.4. Ensure Accessibility Compliance

Use Case Title	Ensure Accessibility Compliance
Use Case ID	UC-AP-04
Actions	• Admin open "Accessibility Compliance" section • System shows list of issues or missing accessibility features • Admin reviews problem areas • Takes steps to correct issues
Description	Admin checks that surveys follow accessibility rules (screen reader, support, alt-text, color contrast etc.).
Alternative Paths	If no issues found, system shows "All Accessible" message
Pre-Conditions	• Survey should exist • Accessibility rules must be defined in system
Post-Conditions	Accessibility issues reviewed or fixed
Author	BC220203921
Exceptions	• System fails to scan accessibility issues • Accessibility rules not updated

3.7. Form Creator Panel

3.7.1. Create Survey

Use Case Title	Create Survey
Use Case ID	UC-FC-01
Actions	• User logs in to the system • Navigates to "Create Survey" page • Enter title and description • Saves the new survey
Description	The Form Creator starts a new survey by entering its basic details
Alternative Paths	• User cancels the process • User leaves required field empty (system show validation message)
Pre-Conditions	• User must be logged in • User must have Form Creator role
Post-Conditions	New empty survey is created and stored in system
Author	BC220203921
Exceptions	• System error while saving data

3.7.2. Add Questions

Use Case Title	Add Questions
Use Case ID	UC-FC-02
Actions	• User opens an existing survey • Clicks "Add Question" • Selects Question type (MCQ, text, rating) • Enter Question and options • Saves the Question
Description	Allows Form Creator to add different question types to a survey.
Alternative Paths	• User edits an existing question • User deletes an added question
Pre-Conditions	A survey must already exist
Post-Conditions	New question is added to survey
Author	BC220203921
Exceptions	• Invalid question format • System unable to save question

3.7.3. Apply Accessibility Settings

Use Case Title	Apply Accessibility Settings
Use Case ID	UC-FC-03
Actions	• User opens accessibility settings • Enable accessibility features like high contrast, large font, alt-text, screen reader support) • Saves changes
Description	Allows Form Creator to enable accessibility features for survey.
Alternative Paths	• User only applies some settings • User reset settings to default
Pre-Conditions	Survey must be created
Post-Conditions	Survey become accessible for all users
Author	BC220203921
Exceptions	• Missing alt-text for images • System fails to update settings

3.7.4. Publish Survey

Use Case Title	Publish Survey
Use Case ID	UC-FC-04
Actions	• User review completed survey • Clicks “Publish” • System generates shareable link/code
Description	The survey becomes live so participants can respond.
Alternative Paths	• User schedules survey for future publish time
Pre-Conditions	• Survey must be completed • Required questions must be added
Post-Conditions	Survey becomes publicly accessible
Author	BC220203921
Exceptions	• Missing necessary details • Publishing service error

3.7.5. View Responses

Use Case Title	View Responses
Use Case ID	UC-FC-05
Actions	• User opens survey dashboard • Clicks “View Responses” • System displays all collected responses
Description	Allows Form Creator to see responses submitted by participants.
Alternative Paths	• User filters responses by date • User downloads responses summary
Pre-Conditions	• A survey must be published • Response data must exist
Post-Conditions	Responses viewed by user
Author	BC220203921
Exceptions	• No Responses availble • System fails to load data

3.7.6. Export Reports

Use Case Title	Export Reports
Use Case ID	UC-FC-06
Actions	• User opens report section • Select format (PDF/Excel) • Click “Export” • System Generates file
Description	Form Creator downloads survey report in PDF/Excel format.
Alternative Paths	• User exports only questions summary • User exports filtered data
Pre-Conditions	Survey must have responses
Post-Conditions	Report is downloaded
Author	BC220203921
Exceptions	• File generation failed • Slow Network

3.8. Respondent Panel

3.8.1. Fill Survey

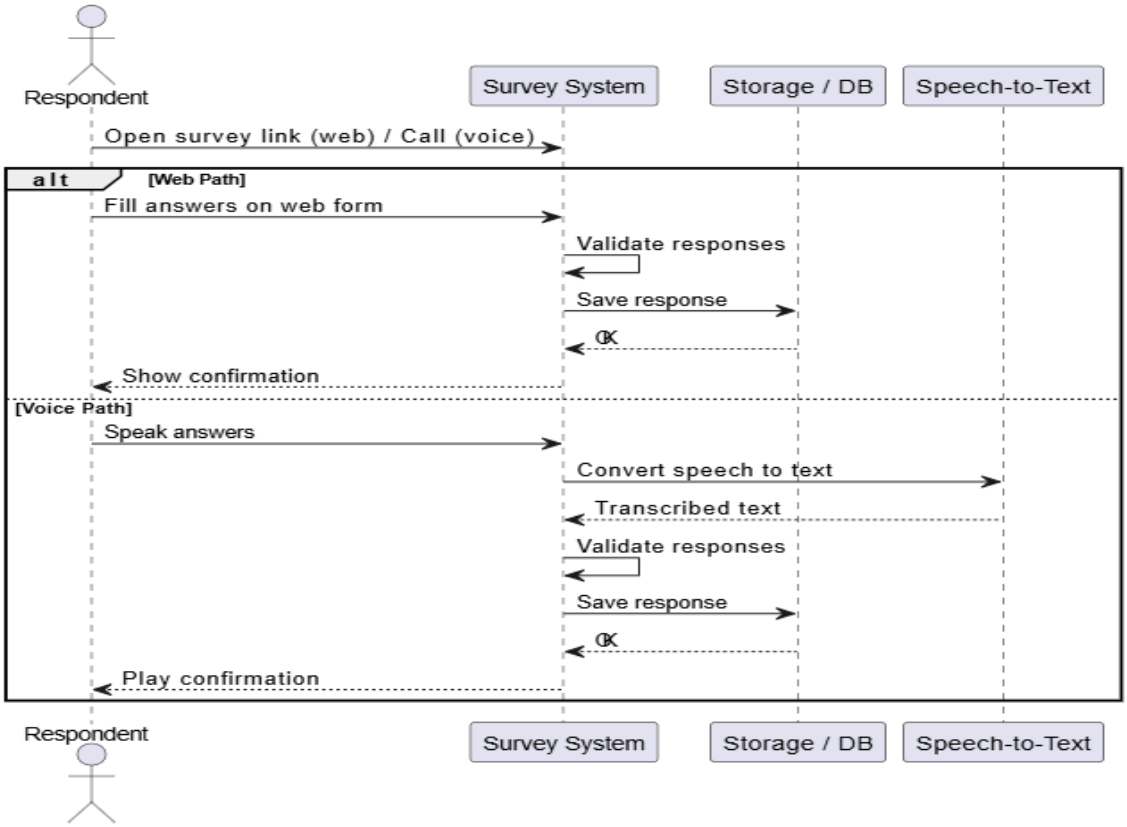
Use Case Title	Fill Survey
Use Case ID	RUC-01
Actions	• Respondent opens the survey link. • System loads the survey form. • Respondent read each question • Respondent selects answers / types responses. • System autosave progress.
Description	This use case describes how a respondent opens the survey link and fills the required questions.
Alternative Paths	• If an answer is missing, system highlights required fields. • If the internet is slow, system loads form in basic mode.
Pre-Conditions	• Respondent must have the survey link • Survey must be active and availble
Post-Conditions	• Survey is filled but not yet submitted • System stores temporary answers
Author	BC220203921
Exceptions	• Survey link invalid or expired • System fails to load questions due to server error.

3.8.2. Submit Response

Use Case Title	Submit Response
Use Case ID	RUC-02
Actions	• Respondent review all answers. • Respondent clicks **"Submit"** button. • System validates all mandatory questions • System saves the final response • System shows confirmation message.
Description	This use case explains how the respondent submit the completed survey to the system.
Alternative Paths	• If required answers are missing, system asks respondent to complete them. • System may retry submission if internet is unstable.
Pre-Conditions	• Respondent must have filled the survey. • Survey must still be open for response submission.
Post-Conditions	• Response is saved in the database. • Respondent cannot edit responses after submission.
Author	BC220203921
Exceptions	• System error during submission.

	• Internet disconnects before submission final save. • Duplicate submission attempt.

3.9. Sequence Diagram — Submit Response



Explanation:

This sequence diagram shows how a respondent submits their surveys. There are three main ways to submit the survey:

- **Web Path:** The respondent fills the survey using the web form. The system checks the answers, saves them into the database, and shows a confirmation message to the user.
- **Voice Path:** The respondent answers using their voice. The speech-to-text module converts it into text, the system validates and saves it in the database, and then plays a confirmation audio for the user.

4. Adopted Methodology:

For **Access Form Management System,** we adopted a **Hybrid Software Development Model**, which is the combination of Waterfall Model, the Spiral Model and the VU Process Model. We choose this model because our project contains form creation, user access, validation accessibility checking and data handling in which only one SDLC model is not enough.

This hybrid model provides combination of structured development (from waterfall model), Risk handling (from spiral method), and Continuous Validation (from VU Process Model).

4.1. Waterfall Model

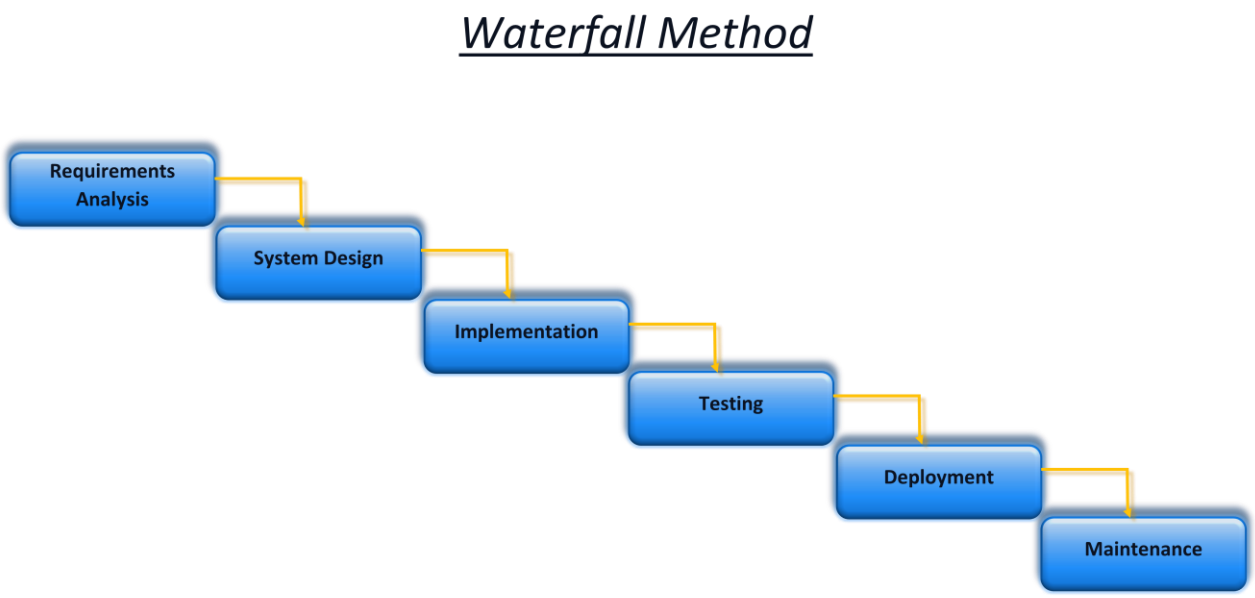
The Waterfall Model is one of the oldest and simplest development models. In the model, the development process complete linearly, in this each phase must be fully completed before moving to the next phase. This is suitable for our project for the following reasons:

- ✓ Our project has clear requirements such as creating forms, managing user roles, submitting surveys, and generating dashboards and reports
- ✓ Documentation is very important in university projects, and the Waterfall Model fits perfectly for this.
- ✓ This project needs a step-by-step flow: first documentation, then design, and finally coding. This matches the Waterfall Model approach.

4.1.1. <u>Waterfall Phases in our Project</u>

- **Requirements Analysis:** All requirements of Access Form are collected and Documented.
- **System Design:** We prepared use case diagrams, Data flow diagrams (DFD's), Entity Relationship Diagrams (ERD) and interface Designs for the system.
- **Implementation:** Module like Admin Panel, Form Creator Panel, Form Submission Panel and Accessibility rules were developed.
- **Testing:** The created forms were tested for validation, correctness, user roles and accessibilities.
- **Deployment & Maintenance:** The final system is prepared for end-user and maintained based on future updates.

4.1.2. Diagram - Waterfall Model



4.2. Spiral Model

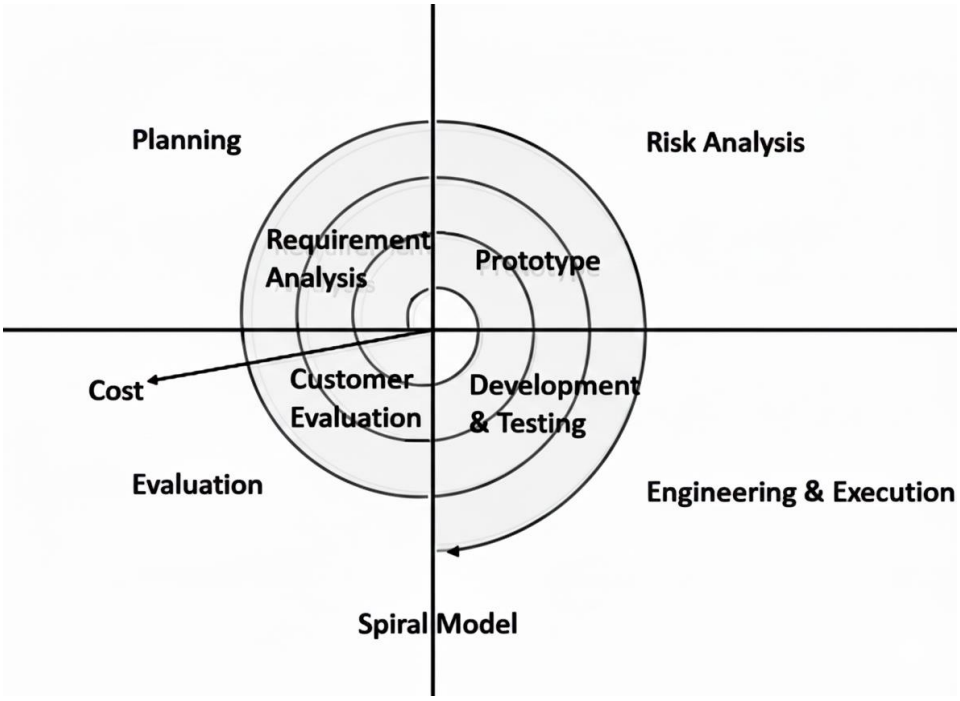
This model focuses on identifying and reducing risk through repeated cycles. As in our project contains multiple points where user input errors, accessibility issues and validation problem may occur, we included spiral elements to reduce development risks. The reason why spiral suits our project is:

- ✓ Our form creation module is risk-based (wrong form structure, user validation errors)
- ✓ Accessibility rules must be validated multiple times.
- ✓ Prototype-based checking is needed in the form creation module
- ✓ Repetitions helps us to improve the system before deployment.

4.2.1. Spiral Model in our Project

- **Planning:** We planned each module such as login, Admin Panel, Form Creation, Submission and Reports.
- **Risk Analysis:** Accessibility issues, incorrect field validation, and wrong form logic were identified and resolved.
- **Engineering & Execution:** The form builder was tested using small prototypes before final development
- **Evaluation:** Errors were fixed based on feedback and improvements were made before moving to the next cycle.

4.2.2. <u>Diagram – Spiral Model</u>



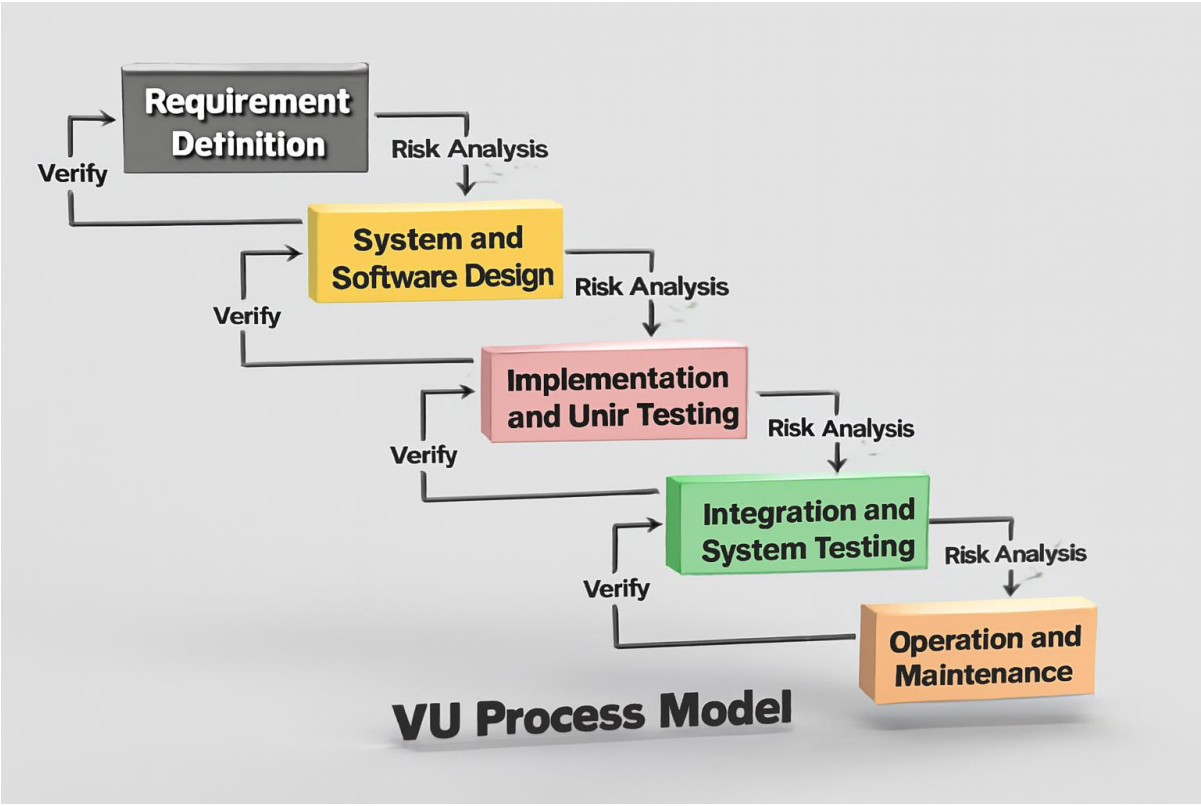
4.3. VU Process Model

The VU Process Model is used by Virtual University to ensure that each step of development is properly validated. It works well with hybrid development, and it focuses on continuously validating requirements, design and implementation. We use this model because our project needs continuous Validations at each stage.

4.3.1. VU Process Model in our Project

- **Requirements:** All form fields, user roles, submission rules, and accessibility constraints were validated before designing the system.
- **Design:** Use case diagrams, DFD's, activity diagrams and layout structured were validated to ensure correctness.
- **Implementation:** The implementation modules such as login, Admin Panel, Form Creator and Submission system were Validated through testing.

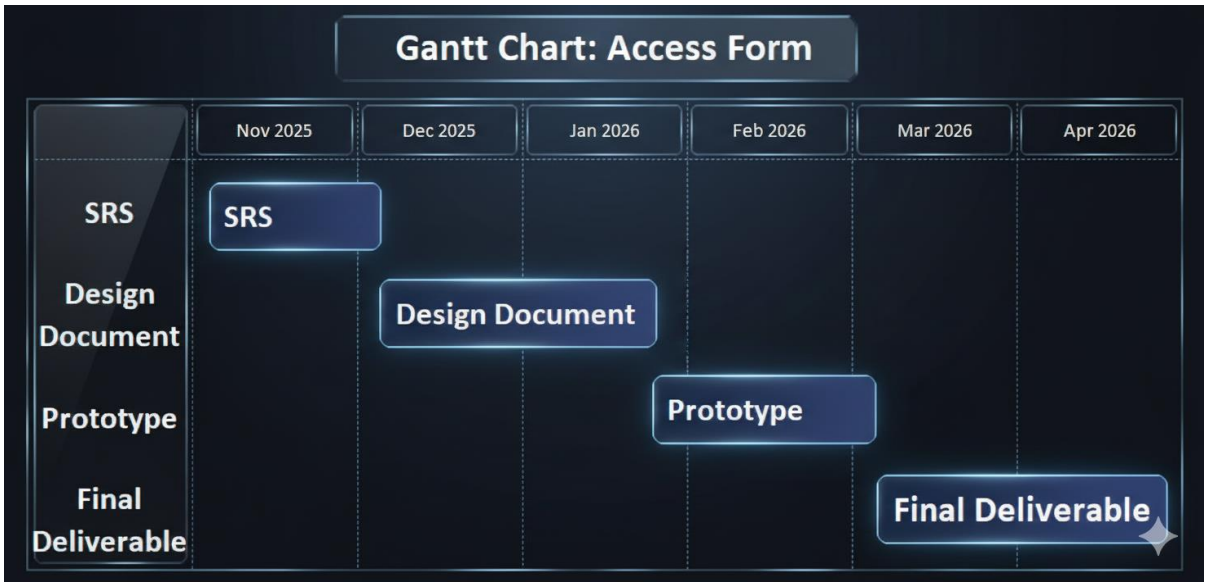
4.3.2. Diagram – VU Process Model



4.4. Why Hybrid Model is Best for our project

We selected the Hybrid Model because it combines the strength of the Waterfall, Spiral and VU Process Models in a balanced way. There are some main features of all model which are used in this project.

- The Waterfall Model supports our project by providing proper documentation, a clear development sequence and a simple structure that is suitable for projects.
- The **Spiral Model** is useful because it helps identify validations and accessibility risks at an early stage. It also allows us to create prototypes of the form builder and refine the system through repetitions.
- The **Vu Process Model** ensure continuous validations throughout the development phases. It Improves the quality of the project and follows the recommended process pattern of the Virtual University.

Overall, our Access Form Management system includes multiple components such as admin functions, form creation, user roles, validation, accessibility roles and response submission. Because this project needs both a structured flow (Waterfall) and flexibility for improvements (Spiral + VU), so that the hybrid model is the most suitable choice.

5. Work Plan